

Outline

- ▶ Basic concepts of association rule learning
- ▶ Apriori algorithm
- ▶ FP-Growth Algorithm
- ▶ Finding interesting rules

Two Problems with Association Rule Mining

▶ Quantity problem

- ▶ Too many rules can be generated
 - ▶ Given a dataset, the number of rules generated depends on the support and confidence thresholds.
 - ▶ If the support threshold is high, a small number of rules are generated. But some interesting rules are missed.
 - ▶ If the support threshold is low, a huge number of rules are generated.

▶ Quality problem

- ▶ Not all the generated rules are interesting

Number of Generated Patterns versus Support Threshold (An Example)

Support threshold	0.02	0.01	0.008	0.005	0.003	0.0028	0.0025	0.002	0.001
Num. of rules (conf. thres.=0.5)	2	14	39	88	723	4,556	74,565	4,800,070	$>10^9$
Num. of rules (conf. thres.=0.8)	1	7	17	38	591	4,172	65,615	3,584,339	$>10^9$

Number of sessions (transactions): 30586

Number of objects (items): 38679

Solutions to the Problems

- ▶ Using interestingness measures to rank and remove rules
 - ▶ Remove misleading associations and find correlation rules
 - ▶ Prune patterns using other interestingness measures
- ▶ Constraint-based data mining
 - ▶ Applying constraints in the mining process so the search can be more focused and the results are more useful.
- ▶ Finding only *maximum* or *closed* frequent patterns
 - ▶ Other frequent patterns can be generated from them
- ▶ Using rule structures
 - ▶ Eliminate structurally and semantically redundant rules.
 - ▶ Group or summarize related rules

Selected Solutions to the Problems

- ▶ *Using interestingness measures to remove or rank rules*
 - ▶ *Remove misleading associations and find correlation rules*
 - ▶ Prune patterns using other interestingness measures
- ▶ Constraint-based data mining
 - ▶ Applying constraints in the mining process so the search can be more focused.

Misleading Association Rules

	Basketball	Not basketball	Sum (row)
Cereal	2000	1750	3750
Not cereal	1000	250	1250
Sum(col.)	3000	2000	5000

- ▶ *play basketball* $\Rightarrow$ *eat cereal* [40%, 66.7%] **is misleading**
 - ▶ The overall percentage of students eating cereal is 75% which is higher than 66.7%
- ▶ *play basketball* $\Rightarrow$ *not eat cereal* [20%, 33.3%] is more positively correlated, although with lower support and confidence

Association $\neq$ Positive Correlation

Interestingness Measure: Correlation

- ▶ Correlation between two events A and B

- ▶ If $P(A|B) > P(A)$, A and B are *positively correlated*.

Note: $P(A | B) > P(A) \Leftrightarrow P(B | A) > P(B) \Leftrightarrow P(A B) > P(A)P(B)$

- ▶ If $P(A|B) < P(A)$, A and B are *negatively correlated*.

Note: $P(A | B) < P(A) \Leftrightarrow P(B | A) < P(B) \Leftrightarrow P(A B) < P(A)P(B)$

- ▶ If $P(A|B) = P(A)$, A and B are *independent*.

Note: $P(A | B) = P(A) \Leftrightarrow P(B | A) = P(B) \Leftrightarrow P(A B) = P(A)P(B)$

- ▶ A measure of correlation (called **lift**)

$$corr_{A,B} = \frac{P(AB)}{P(A)P(B)}$$

Pruning Misleading Rules (Keep Correlation Rules)

- ▶ A measure of correlation (**lift**) for rule $A \rightarrow B$

$$\text{lift}(A \rightarrow B) = \frac{P(AB)}{P(A)P(B)}$$

- ▶ Rules whose lift ≤ 1 is **misleading**, which should be removed

- ▶ E.g. the following rule:

play basketball $\Rightarrow$ eat cereal [40%, 66.7%]

should be removed because its lift is 0.89

Solutions to the Problems

- ▶ *Using interestingness measures to remove or rank rules*
 - ▶ Remove misleading associations and find correlation rules
 - ▶ *Prune patterns using other interestingness measures*
- ▶ Constraint-based data mining
 - ▶ Applying constraints in the mining process so the search can be more focused.

Many interestingness measures for $A \rightarrow B$

symbol	measure	range	formula
ϕ	ϕ -coefficient	-1 ... 1	$\frac{P(A,B) - P(A)P(B)}{\sqrt{P(A)P(B)(1-P(A))(1-P(B))}}$
Q	Yule's Q	-1 ... 1	$\frac{P(A,B)P(\bar{A},\bar{B}) - P(A,\bar{B})P(\bar{A},B)}{P(A,B)P(\bar{A},\bar{B}) + P(A,\bar{B})P(\bar{A},B)}$
Y	Yule's Y	-1 ... 1	$\frac{\sqrt{P(A,B)P(\bar{A},\bar{B})} - \sqrt{P(A,\bar{B})P(\bar{A},B)}}{\sqrt{P(A,B)P(\bar{A},\bar{B})} + \sqrt{P(A,\bar{B})P(\bar{A},B)}}$
k	Cohen's	-1 ... 1	$\frac{P(A,B) + P(\bar{A},\bar{B}) - P(A)P(B) - P(\bar{A})P(\bar{B})}{1 - P(A)P(B) - P(\bar{A})P(\bar{B})}$
PS	Piatetsky-Shapiro's	-0.25 ... 0.25	$P(A,B) - P(A)P(B)$
F	Certainty factor	-1 ... 1	$\max\left(\frac{P(B A) - P(B)}{1 - P(B)}, \frac{P(A B) - P(A)}{1 - P(A)}\right)$
AV	added value	-0.5 ... 1	$\max(P(B A) - P(B), P(A B) - P(A))$
K	Klogsen's Q	-0.33 ... 0.38	$\sqrt{P(A,B)} \max(P(B A) - P(B), P(A B) - P(A))$
g	Goodman-kruskal's	0 ... 1	$\frac{\sum_j \max_k P(A_j, B_k) + \sum_k \max_j P(A_j, B_k) - \max_j P(A_j) - \max_k P(B_k)}{2 - \max_j P(A_j) - \max_k P(B_k)}$
M	Mutual Information	0 ... 1	$\frac{\sum_i \sum_j P(A_i, B_j) \log \frac{P(A_i, B_j)}{P(A_i)P(B_j)}}{\min(-\sum_i P(A_i) \log P(A_i) \log P(A_i), -\sum_i P(B_i) \log P(B_i) \log P(B_i))}$
J	J-Measure	0 ... 1	$\max(P(A,B) \log\left(\frac{P(B A)}{P(B)}\right) + P(\bar{A}\bar{B}) \log\left(\frac{P(\bar{B} \bar{A})}{P(\bar{B})}\right), \\ P(A,B) \log\left(\frac{P(A B)}{P(A)}\right) + P(\bar{A}\bar{B}) \log\left(\frac{P(\bar{A} \bar{B})}{P(\bar{A})}\right))$
G	Gini index	0 ... 1	$\max(P(A)[P(B A)^2 + P(\bar{B} \bar{A})^2] + P(\bar{A})[P(B \bar{A})^2 + P(\bar{B} \bar{A})^2] - P(B)^2 - P(\bar{B})^2, \\ P(B)[P(A B)^2 + P(\bar{A} B)^2] + P(\bar{B})[P(A \bar{B})^2 + P(\bar{A} \bar{B})^2] - P(A)^2 - P(\bar{A})^2)$
s	support	0 ... 1	$P(A,B)$
c	confidence	0 ... 1	$\max(P(B A), P(A B))$
L	Laplace	0 ... 1	$\max\left(\frac{NP(A,B)+1}{NP(A)+2}, \frac{NP(A,B)+1}{NP(B)+2}\right)$
IS	Cosine	0 ... 1	$\frac{P(A,B)}{\sqrt{P(A)P(B)}}$
γ	coherence(Jaccard)	0 ... 1	$\frac{P(A,B)}{P(A)+P(\bar{B})-P(A,B)}$
α	all_confidence	0 ... 1	$\frac{P(A,B)}{\max(P(A), P(B))}$
o	odds ratio	0 ... ∞	$\frac{P(A,B)P(\bar{A},\bar{B})}{P(\bar{A},B)P(A,\bar{B})}$
V	Conviction	0.5 ... ∞	$\max\left(\frac{P(A)P(\bar{B})}{P(A\bar{B})}, \frac{P(B)P(\bar{A})}{P(B\bar{A})}\right)$
λ	lift	0 ... ∞	$\frac{P(A,B)}{P(A)P(B)}$
S	Collective strength	0 ... ∞	$\frac{P(A,B) + P(\bar{A}\bar{B})}{P(A)P(B) + P(\bar{A})P(\bar{B})} \times \frac{1 - P(A)P(B) - P(\bar{A})P(\bar{B})}{1 - P(A,B) - P(\bar{A}\bar{B})}$
χ^2	χ^2	0 ... ∞	$\sum_i \frac{(P(A_i) - E_i)^2}{E_i}$

Pruning rules with interestingness measure

- ▶ Choose a measure in your belief to assess the significance of a rule $A \rightarrow B$.
- ▶ Rank the rules according to their interestingness value.
- ▶ Remove rules with small interestingness values

Solutions to the Problems

- ▶ Using interestingness measures to remove or rank rules
 - ▶ Remove misleading associations and find correlation rules
 - ▶ Prune patterns using other interestingness measures
- ▶ *Constraint-based data mining*
 - ▶ *Applying constraints in the mining process so the search can be more focused.*

Constrain-based Frequent Pattern and Association Rule Mining

- ▶ Mining frequent patterns or rules with constraint C
 - ▶ find all frequent patterns satisfying not only min_sup, but also constraint C
 - ▶ find all association rules satisfying not only min_sup and min_conf, but also constraint C
- ▶ Examples of Constraints
 - ▶ $? \rightarrow$ *a particular product*
 - ▶ *a particular product* $\rightarrow$ $?$
 - ▶ small sales (price $<$ \$10) triggers big sales (sum $>$ \$200)

Constrain-based Frequent Pattern Mining (Cont'd)

- ▶ A naïve solution
 - ▶ Testing frequent patterns on C as a post-processing process
- ▶ Some constraints can be incorporated into the mining process to improve the efficiency
- ▶ More efficient approaches
 - ▶ Analyze the properties of constraints comprehensively
 - ▶ Push the constraint as deeply as possible inside the frequent pattern mining
 - ▶ Example: find all frequent itemsets containing item “b”
 - ▶ Can be easily incorporated in FP-growth.
 - ▶ What about Apriori?

Types of Constraints

- ▶ Anti-monotonic constraints

- ▶ An itemset S satisfies the constraint, so does any of its subset (That is, S violates the constraint, so does any of its superset).

- ▶ Monotonic constraints

- ▶ An itemset S satisfies the constraint, so does any of its superset

- ▶ Examples

- ▶ Sum of the prices of items in $S \leq 100$ is anti-monotonic
 - ▶ Maximum price in $S \leq 15$ is anti-monotonic
 - ▶ Sum of the prices of items in $S \geq 100$ is monotonic
 - ▶ Minimum price in $S \leq 15$ is monotonic

How to Use Antimonotonic or Monotonic Constraints in Mining

- ▶ Anti-monotonic constraints
 - ▶ In Apriori:
 - ▶ Use it to prune candidates in each iteration
 - ▶ In FP-growth
 - ▶ Use it to stop growing a pattern
- ▶ Monotonic constraints
 - ▶ If an itemset satisfies a monotonic constraint, no need to check its supersets on the constraint
 - ▶ Only checks their support

Types of Constraints (Cont'd)

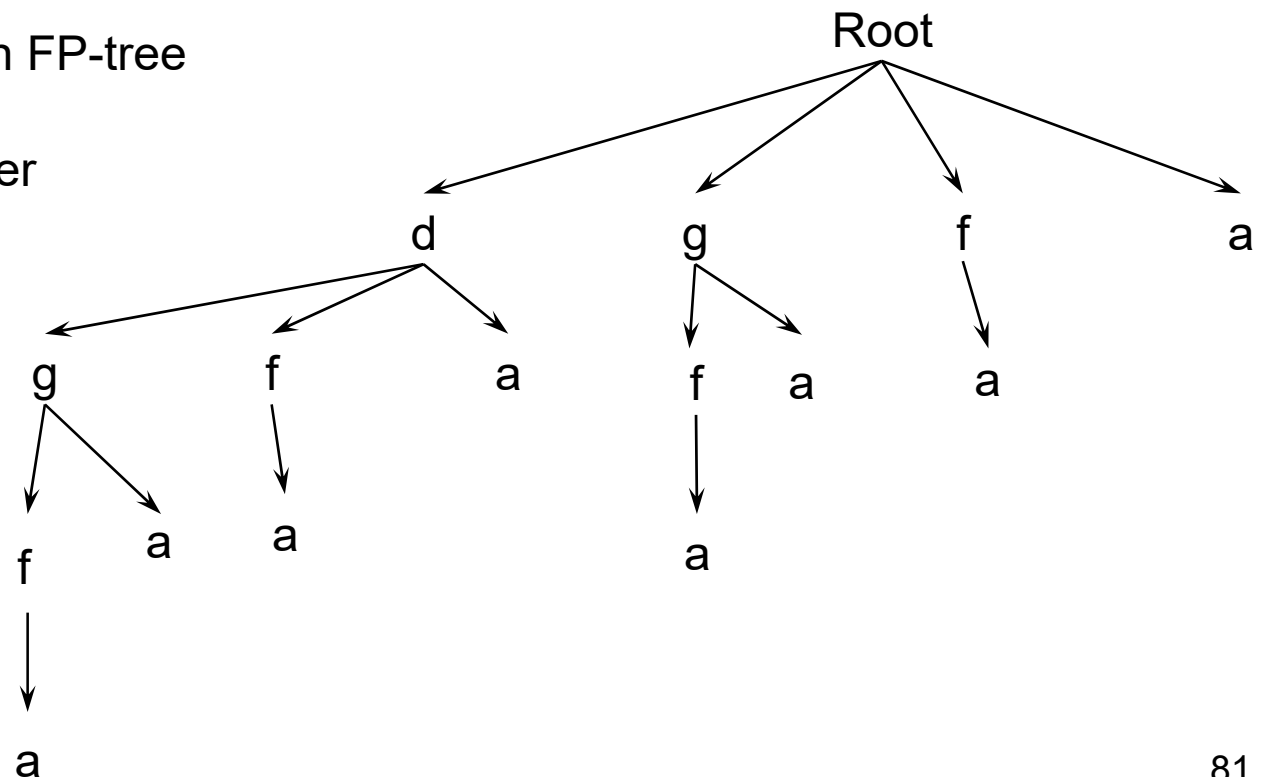
- ▶ Convertible constraints
 - ▶ Some constraints are not anti-monotonic or monotonic
 - ▶ But can be converted to anti-monotonic or monotonic by properly ordering items
- ▶ Example of convertible constraint:
 - ▶ Average price of the items in $S \geq 25$
 - ▶ Order items in price-descending order
 - ▶ $\langle a, f, g, d, b, h, c, e \rangle$
 - ▶ If an itemset afb violates C
 - ▶ So does $afbh$, afb^*
 - ▶ It becomes anti-monotone!

Example of Convertible Constraints

- ▶ Convertible constraint:
 - ▶ Average price of the items in $S \geq 25$
- ▶ Price-descending order of items: a, f, g, d

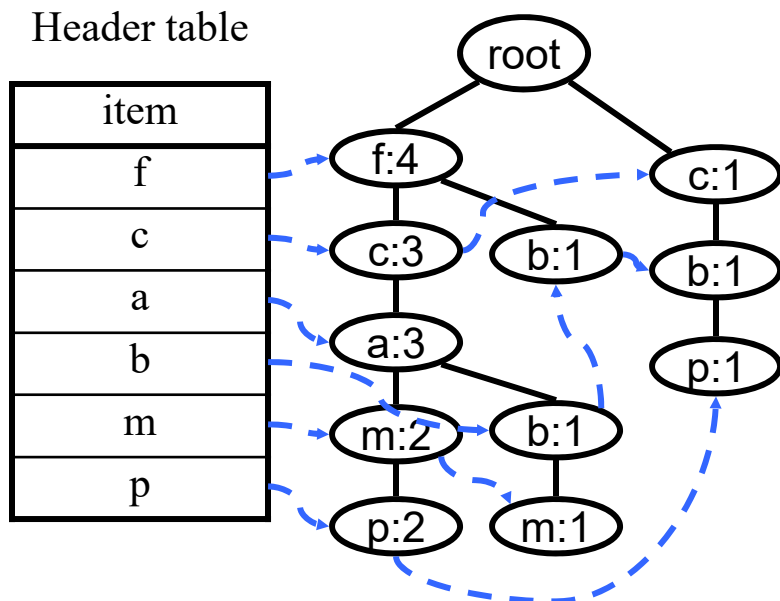
How would you build an FP-tree to do such a search?

- Price-ascending order



FP-Growth (Recap)

- ▶ Build the tree in item-frequency-descending order (**f-list**: **f-c-a-b-m-p**)
- ▶ Grow frequent patterns in frequency-ascending order:
 - ▶ p->cp
 - ▶ m->am -> cam->fcam



Conditional pattern bases

<u>item</u>	<u>cond. pattern base</u>
<i>c</i>	<i>f:3</i>
<i>a</i>	<i>fc:3</i>
<i>b</i>	<i>fca:1, f:1, c:1</i>
<i>m</i>	<i>fca:2, fcab:1</i>
<i>p</i>	<i>fcam:2, cb:1</i>

Related Topics

- ▶ Contrast pattern mining
 - ▶ Finding patterns and models contrasting two or more classes or conditions
 - ▶ Contrasting groups:
 - ▶ Objects at different time periods
 - ▶ Objects at different spatial locations
 - ▶ Objects across different classes.
 - ▶ Measures for measuring the difference
 - ▶ Frequent/infrequent
 - ▶ Frequency ratio
 - ▶ Odds ratio, etc.
 - ▶ A challenge: need to find the supports of infrequent itemsets in a group.

Related Topics

- ▶ Mining frequent patterns over data streams
 - ▶ A continuous flow of data generated often at high-speed in a dynamic, time-changing environment
 - ▶ Memory and disk are limited to hold all the data
 - ▶ Processing time may be limited by the rate of arrival of instances
 - ▶ **One scan of data set** is required for online mining
 - ▶ Pattern changes over time
 - ▶ Incremental learning
 - ▶ Change detection
 - ▶ etc

Related Topics

▶ Mining high utility patterns

▶ Consider

- ▶ the quantity $q(i, T_j)$ of an item i in a transaction T_j
- ▶ the value (e.g., price $p(i)$) of an item i

▶ Utility of an item i in a transaction T_j :

$$u(i, T_j) = q(i, T_j) \times p(i)$$

▶ Utility of an itemset X in a transaction T_j :

$$u(X, T_j) = \sum_{X \subseteq T_j, i \in X} u(i, T_j)$$

▶ Utility of an itemset X in a dataset D :

$$u(X, D) = \sum_{X \subseteq T_j, T_j \in D} u(X, T_j)$$

- ▶ **High utility pattern**: itemsets whose utility in the dataset is no less than a minimum utility threshold

Related Topics

- ▶ Mining high utility patterns (*cont'd*)
 - ▶ Challenge: utility does not have the downward closure property. That is,

The utility of a subset /superset of a set S may be smaller or larger than the utility of S
 - ▶ This means we cannot use Apriori or FP-growth to find high utility patterns directly since
 - ▶ the two algorithms use the downward closure property of support to cut down the search space

Related Topics

- ▶ Mining high utility patterns (*cont'd*)
 - ▶ Solution:
 - ▶ Use a utility function f that is:
 - ▶ an upper bound of the true utility of a pattern X , i.e., $u(X, T_j) \leq f(X, T_j)$
 - ▶ with downward closure property
 - to generate candidates first
 - ▶ Then scan DB to find high utility patterns from the set of candidates

Next Topic

- ▶ Sequential pattern mining (papers on the supplementary reading list)